

State-Dependent Overcompensation and Piecewise Temperature Response: New Observations from 3D NAND Cross-Temperature Characterization

Yicheng Ke, Minghui Dong, Linlin Cai, Wangyong Chen*

School of Microelectronics Science and Technology, Sun Yat-sen University, Zhuhai 519082, China

*Email: chenwangy@mail.sysu.edu.cn

Abstract—On-chip temperature compensation (TC) is essential for 3D NAND reliability, yet its efficacy across wide temperature ranges and high program states remains underexplored. This work characterizes the threshold voltage (V_{TH}) distributions of TLC states (P1–P7) from -30°C to 125°C , comparing TC-enabled and TC-disabled behaviors. While TC effectively suppresses drift within the nominal window (0°C – 70°C), we report a distinct overcompensation phenomenon in high program states (P5–P7) at elevated temperatures, leading to an abnormal mean-drift inversion and degraded inter-state margins. This anomaly arises from a mismatch between the linear TC scheme and the intrinsic piecewise temperature coefficient (T_{co}) of the device. TCAD simulations confirm that this piecewise behavior stems from a mechanism transition: trap-stabilized charge transport dominates at low temperatures, whereas Poole–Frenkel emission and mobility degradation amplify sensitivity at high temperatures. Our findings highlight the limitations of state-insensitive TC and motivate state-aware calibration to ensure robustness in extreme thermal environments.

Index Terms—3D NAND flash, temperature compensation, program states, over-compensation, NT_{co} .

I. INTRODUCTION

Three-dimensional (3D) NAND flash memory has established itself as the dominant nonvolatile storage solution, driven by its superior density and cost-efficiency [1]. As the technology scales aggressively in layer count and wordline pitch, ensuring read reliability becomes increasingly challenging. The integrity of data retrieval is fundamentally dictated by the tightness of threshold voltage (V_{TH}) distributions and the guard-band margins between multi-level states [2]–[7]. Among various reliability detractors, temperature variation poses a critical threat. It induces significant V_{TH} drift and distribution distortion by altering charge trapping dynamics and modifying read sensing conditions, thereby jeopardizing the sensing accuracy of multi-level cells [2], [4]–[8].

To mitigate these thermal effects, commercial 3D NAND products universally integrate on-chip temperature compensation (TC). This mechanism dynamically adjusts sensing reference voltages based on pre-characterized temperature coefficients to stabilize the read window [9], [10]. However, conventional TC schemes typically rely on a simplified, state-independent linear correction model. This one-size-fits-all approach assumes a uniform thermal response across all program states, ignoring the complex, state-dependent physics of charge trap devices. As program levels increase, the stored

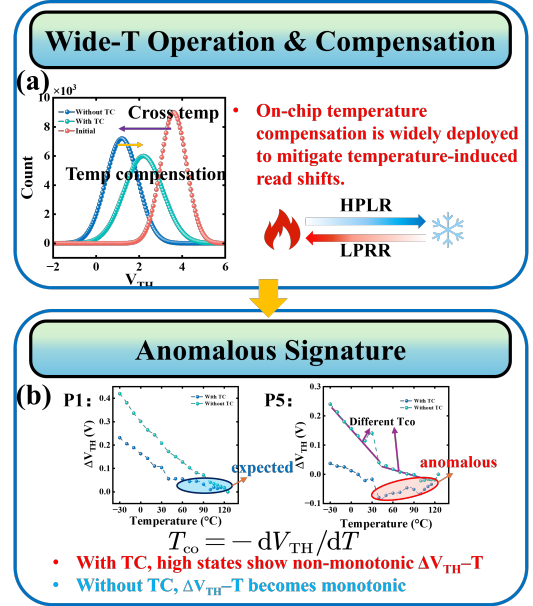


Fig. 1. Distribution-centric evaluation of on-chip TC in charge-trap 3D NAND. (a) TC mitigates temperature-induced V_{TH} drift while preserving inter-state separation. (b) With TC enabled, high program states exhibit a nonmonotonic response; without TC, the response remains monotonic in $\Delta V_{TH}-T$. HPLR and LPHR denote high temperature program low temperature read and low temperature program high temperature read, respectively. $\Delta V_{TH}(T) = V_{TH}(T) - V_{TH}(T_{prog})$.

charge density and vertical field distribution change significantly, potentially leading to a mismatch between the intrinsic device response and the applied compensation. Consequently, while TC may stabilize nominal states, it risks introducing residual errors or even overcompensating high program states, a phenomenon that degrades inter-state linearity and remains insufficiently understood in current literature. As shown in Fig. 1(a), TC suppresses temperature-induced V_{TH} drift and preserves inter-state separation, whereas Fig. 1(b) shows that high program states exhibit a nonmonotonic $\Delta V_{TH}(T)$ under TC, in contrast to the monotonic temperature dependence observed without TC. This mismatch is further reflected at the distribution level, where TC-enabled and TC-disabled V_{TH} distributions diverge across P1–P7 under cross-temperature program and read conditions, as summarized in Fig. 2.

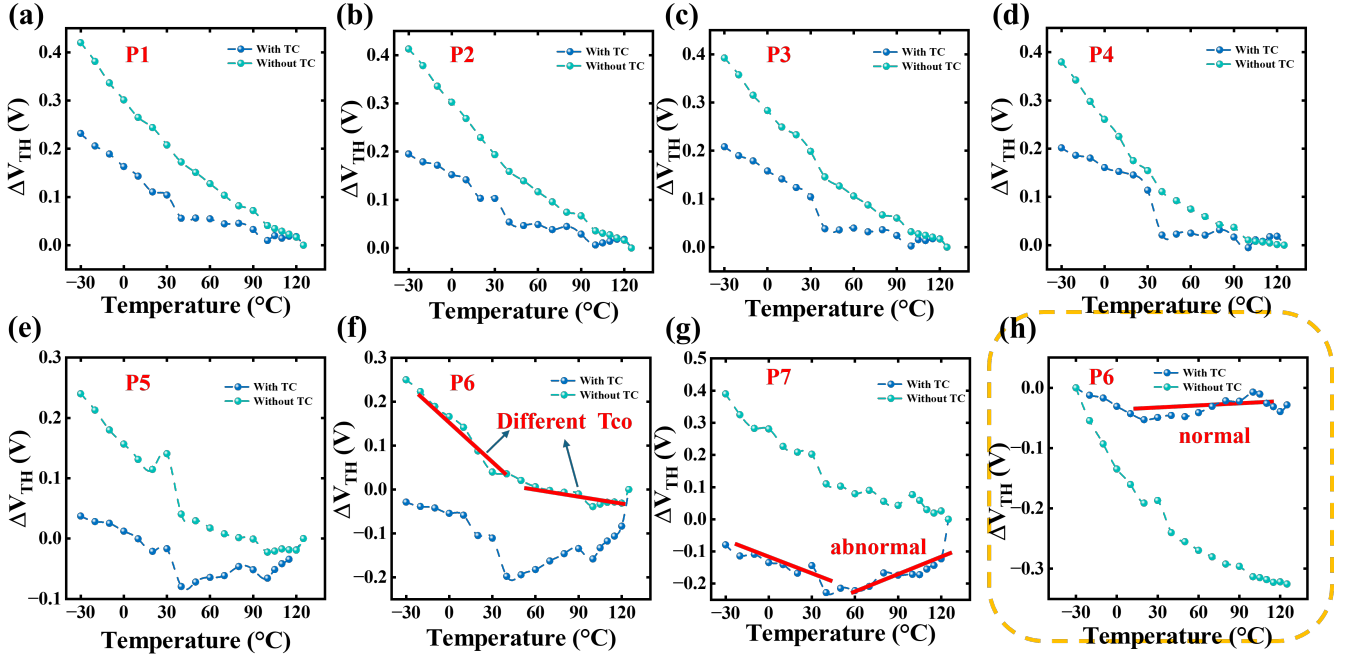


Fig. 2. ΔV_{TH} comparison measured with TC and without TC. (a)–(g) P1–P7 under HPLR. (h) P6 under LPHR.

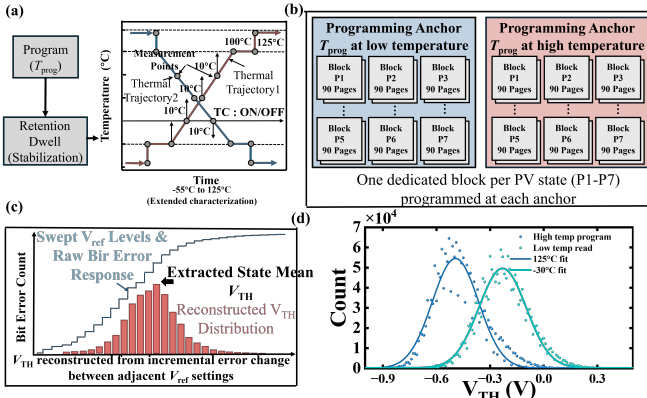


Fig. 3. Experimental flow for distribution extraction and cross temperature evaluation. (a) Temperature sweep strategy and thermal trajectories. (b) Block allocation and state wise programming for P1 to P7. (c) V_{TH} distribution reconstruction using a V_{ref} sweep and raw bit error counting. (d) Example reconstructed V_{TH} distribution for P2 under HPLR.

In this work, we conduct a comprehensive, distribution-level assessment of on-chip TC in charge-trap 3D NAND to expose the limitations of conventional compensation schemes. Unlike prior studies that focus primarily on mean-value trends, we employ a rigorous comparative methodology, tracking V_{TH} evolution for states P1 to P7 under both TC-enabled and TC-disabled conditions along controlled cross-temperature trajectories (-30°C to 125°C). Our key innovation lies in identifying a distinct piecewise temperature dependence and a counter-intuitive mean-drift inversion in high program states when TC is active. By decoupling the intrinsic thermal response from compensation artifacts, we reveal that the high-temperature

anomaly is driven by a mechanism transition involving Poole-Frenkel emission and mobility degradation. These findings provide critical physical insights and motivate the need for next-generation, state-aware TC calibration strategies to ensure robust reliability in extreme thermal environments.

II. CROSS-TEMPERATURE TC SEPARATION EXPERIMENT

The commercial charge-trap 232-layer 3D NAND flash memory chips were characterized in this study. All tested samples were from the same fabrication lot, including three chips with two measured dies per chip. The vendor specifies a nominal operating temperature range of 0°C to 70°C . To capture the full temperature response and its regime transition, and to reveal potential TC mismatch that may not manifest within the nominal window, we extend the characterization to -30°C to 125°C along controlled thermal trajectories. The on-chip TC option was toggled to obtain paired measurements with TC enabled and disabled under identical test flows.

Fig. 3(a) shows the temperature sweep strategy and thermal trajectories. To avoid state mixing, each die used seven dedicated blocks, with one block assigned to each PV state from P1 to P7 as illustrated in Fig. 3(b), and 90 pages were sampled from each state block at each temperature. Two program-temperature anchors are adopted, representing low-temperature and high-temperature programming [5], [6]. After programming, the V_{TH} distributions are measured along two thermal trajectories using a thermal-shock system. A sufficient retention dwell is applied prior to the sweep to suppress early postprogram relaxation [11], [12]. All measurements are taken after temperature stabilization under the same sensing criterion.

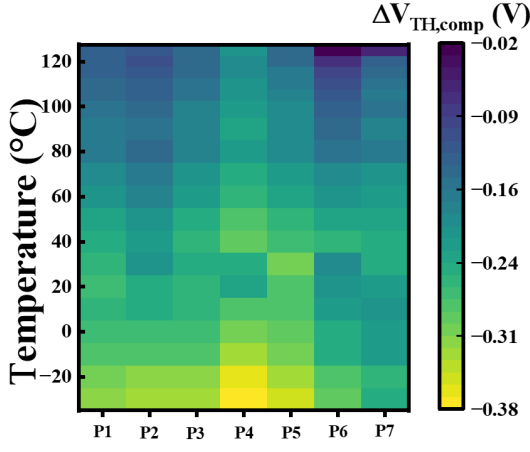


Fig. 4. TC heatmap of effective compensation at the extracted-mean level $\Delta V_{TH,comp}(T, P) = V_{TH}^{WithTC}(T, P) - V_{TH}^{WithoutTC}(T, P)$ versus temperature and program state (P1–P7).

Fig. 3(c) summarizes the V_{TH} extraction. The distribution was reconstructed by sweeping the read reference voltage V_{ref} and recording raw bit error counts for each sampled page. The incremental error change between adjacent V_{ref} settings corresponds to the population of cells whose effective thresholds fall within that interval, enabling a histogram like reconstruction of the V_{TH} distribution [13]. The state mean V_{TH} was then extracted using an identical estimator across all temperatures, states, and conditions with TC and without TC to ensure a fair comparison. The same readout configuration and V_{ref} sweep procedure are used with TC enabled and disabled, so that the extracted V_{TH} is evaluated in the same readout domain. Temperature coefficient (T_{co}) is then obtained from the extracted $V_{TH}(T)$. Fig. 3(d) shows an example P2 distribution under high temperature program and low temperature read.

III. RESULTS AND DISCUSSION

A. Mean Drift Behavior Across Program States With TC and Without TC

Under the HPLR trajectory, high program states exhibit a more pronounced mean-drift inversion at elevated temperatures, as the $\Delta V_{TH}(T)$ trend reverses with increasing temperature. This inversion trend is observed consistently across all tested dies from the three chips and across seven blocks per die. Fig. 2(a)–(g) compare $\Delta V_{TH}(T)$ extracted from the distribution mean with TC enabled and disabled for P1–P7. As shown in Fig. 2(a)–(d), TC effectively stabilizes the mean threshold voltage for lower states (P1–P4), maintaining distribution stability within the nominal operating window (0°C–70°C). In stark contrast, high program states (P5–P7) in Fig. 2(e)–(g) exhibit a distinct anomaly: with TC enabled, the mean response deviates from the monotonic thermal trend observed in the TC-disabled baseline, manifesting as a drift inversion at elevated temperatures. This behavior severely degrades inter-state linearity by compressing the read margins between neighboring high states.

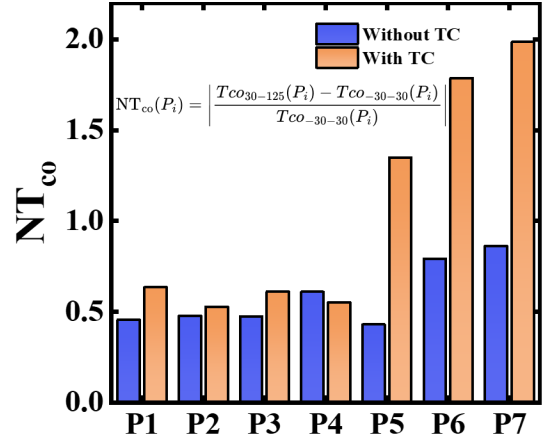


Fig. 5. Normalized T_{co} (NT_{co}) extracted from the segmented T_{co} comparison, showing behaviors with TC and without TC across program states (P1–P7).

Fig. 2(h) presents P6 measured under the LPHR trajectory. In this case, the TC behavior remains well controlled across temperature, indicating that LPHR is more effectively compensated for this state than HPLR at elevated temperatures.

B. State and Temperature Map of TC Effect and Piecewise T_{co}

As shown in Fig. 4, the TC compensation magnitude is primarily governed by temperature and exhibits only weak dependence on program state. A consistent trend is observed from P1 to P7, where the applied correction is small at low temperatures and increases toward higher temperatures, indicating a stronger reference adjustment in the high-temperature region. This weak state dependence becomes problematic at elevated temperature under large stored charge, where the mean response shows clear program-state dependence. As a result, a weakly state dependent correction cannot capture the high-state mean response and tends to overcompensate P5 to P7, contributing to the high-temperature inversion.

To quantify the regime transition, we define the Normalized Temperature Coefficient (NT_{co}) in Eq. (1), which captures the ratio of high-temperature sensitivity to low-temperature sensitivity. As shown in Fig. 5, the NT_{co} trend clarifies how TC reshapes the mean temperature response when the slope is evaluated in a segmented manner.

$$NT_{co}(P_i) = \left| \frac{T_{co30-125}(P_i) - T_{co-30-30}(P_i)}{T_{co-30-30}(P_i)} \right| \quad (1)$$

In the segmented-slope analysis, NT_{co} is consistently larger with TC than without TC, and the difference becomes pronounced in high program states, where NT_{co} exceeds 1. This indicates that TC amplifies the piecewise slope change of $V_{TH}(T)$ in high states. The trend is consistent with the abnormal high-temperature V_{TH} evolution observed with TC and points to overcompensation caused by program-state-dependent mismatch.

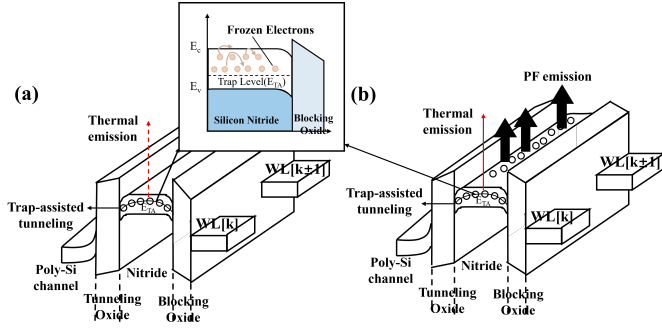


Fig. 6. Proposed dominant mechanisms for V_{TH} temperature dependence in charge-trap 3D NAND. (a) At low temperature, suppressed emission and trap-assisted transport stabilize stored charge. (b) At high temperature, Poole-Frenkel emission under strong vertical field and mobility degradation dominate.

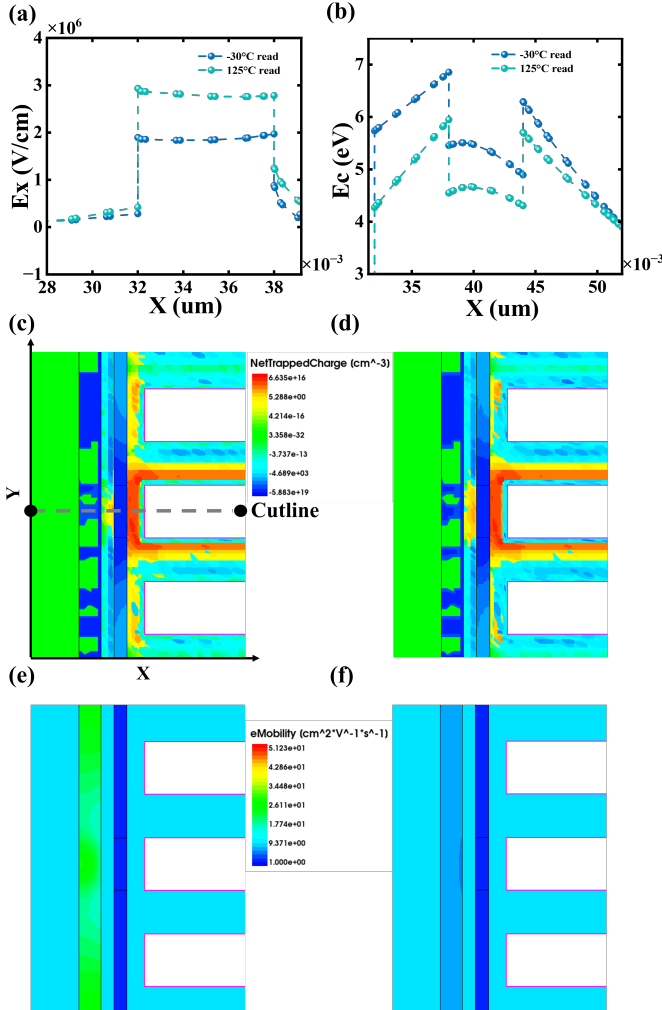


Fig. 7. TCAD support for piecewise T_{CO} . (a) E_x line cuts across the charge-trap stack at -30°C and 125°C . (b) Corresponding E_c line cuts along the same path. (c),(d) NetTrappedCharge distribution in the charge-trap nitride at -30°C and 125°C , respectively. (e),(f) Electron mobility distribution in the channel region at -30°C and 125°C , respectively.

C. Physical Origin of the High Temperature Anomaly in High Program States

To explain the observed piecewise behavior of T_{CO} , we attribute the temperature response of V_{TH} to a transition of the dominant mechanisms from low temperature to high temperature. Fig. 6(a) summarizes the low-temperature regime, where thermally activated emission and trap-assisted transport are strongly suppressed [8], [14], [15]. As a result, shallow-trap charge is more stabilized and the charge-related contribution to V_{TH} becomes weakly temperature dependent, leading to a relatively mild and consistent T_{CO} at low temperature. Importantly, under low-temperature programming, the injected electrons have limited ability to relax into deeper configurations and can remain in a more shallow or non-equilibrium distribution. During the subsequent temperature ramp toward high temperature with TC disabled, this shallow component is rapidly depleted or redistributed, which introduces additional mean drift and yields a larger high-temperature slope than the high-temperature-program case, where the trapped-charge configuration has already relaxed and the mean drift tends to saturate.

At high temperature, the dominant processes shift to those illustrated in Fig. 6(b). Under a strong vertical field, Poole-Frenkel emission is enhanced, effectively lowering the barrier and accelerating charge emission and loss, which increases the temperature sensitivity of the charge-related term in V_{TH} [8], [16], [17]. Furthermore, significant channel mobility degradation at high temperatures, as shown in Fig. 7(f), reduces the cell current. Under a constant-current sensing criterion, this degradation forces an increase in the applied gate voltage to sustain the target current, thereby appearing as an apparent increase in V_{TH} . This effect adds a positive component to the V_{TH} drift, further steepening the high-temperature slope. The combined effects cause T_{CO} at high temperature to deviate from its low-temperature value, resulting in piecewise slopes that are more pronounced in high program states.

These interpretations are supported by Sentaurus TCAD simulations using a 2D cylindrical cross-section of a charge-trap 3D NAND cell. The simulated stack includes a silicon channel, tunnel oxide, charge-trap nitride, and blocking oxide with TiN word-line and select-gate electrodes. Carrier transport was solved with coupled Poisson and drift-diffusion equations using SRH and Auger recombination, a temperature-sensitive mobility model, and trap-assisted charge transport including Poole-Frenkel emission and nonlocal barrier tunneling in the dielectric stack. Cross temperature operation was emulated by ramping the device temperature between anchors with traps active during the transition and frozen during the subsequent hold and read phases under readout-matched bias conditions [18].

Figure 7(a) presents the vertical electric field (E_x) across the charge-trap stack at -30°C and 125°C . At 125°C , the electric field becomes significantly stronger, which is consistent with the expected enhancement of Poole-Frenkel emission under elevated vertical bias. However, despite this increased field,

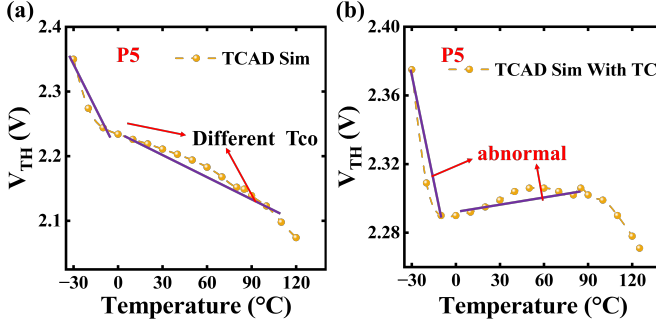


Fig. 8. TCAD validation results. (a) Simulated V_{TH} versus temperature with piecewise slopes. (b) Simulated high-state high-temperature inversion under temperature compensation.

the observed shift in V_{TH} slows down at high temperature, indicating that the charge loss enters a saturation regime. In the early stage, the intensified E_x accelerates the emission of electrons from shallow traps. Once these easily emitted carriers are depleted, the remaining electrons in deeper traps are more difficult to release, and the overall emission rate begins to decrease. As shown in Fig. 7(b), the corresponding conduction band profile (E_c) exhibits only mild variation at 125°C . The band bending is slightly relaxed, suggesting some charge loss, but no substantial flattening occurs. This indicates that a considerable portion of the trapped charge remains even at high temperature, which corroborates the presence of charge-loss saturation.

Figs. 7(c) and 7(d) further support the temperature-driven evolution of trapped charge. Compared with -30°C , the Net-TrappedCharge distribution in the nitride at 125°C becomes more localized. The spatial extent of the trapped-charge cloud shrinks, and the low-density tail is markedly suppressed. This trend indicates preferential emission of electrons residing in shallow or weakly bound traps that primarily populate the dispersed tail region, whereas the remaining charge is dominated by deeper traps that are more difficult to de-trap. Consequently, the charge-loss process approaches a saturation regime, such that further temperature increase yields only limited reduction of trapped charge and progressively suppresses the V_{TH} drift.

Figures 7(e) and 7(f) show the spatial distribution of channel electron mobility at -30°C and 125°C . As expected, the mobility degrades significantly at high temperature, especially near the conduction path. This degradation reduces the channel conductivity under fixed gate bias, effectively increasing the required gate voltage for turn-on. In high program states, where the threshold voltage is already elevated and channel conduction is more sensitive, this mobility-induced degradation further suppresses the cell's ability to conduct.

These results confirm that the mechanism transition is accurately captured in the simulation, particularly the flattening of $V_{TH}(T)$ in high program states due to limited charge emission and mobility degradation at elevated temperature. This physical behavior leads to a segmented temperature response and clearly reveals the mismatch between the actual

device characteristics and conventional TC schemes employing a fixed, state-independent correction slope.

The impact of this mismatch is further demonstrated in Fig. 8. As shown in Fig. 8(a), the simulated V_{TH} curves exhibit piecewise temperature coefficients, which are not reflected in the linear TC model. When the fixed-slope compensation is applied, Fig. 8(b) shows that high program states receive excessive correction at high temperature, resulting in a reversal of the expected V_{TH} drift direction. This overcompensation-induced inversion matches experimental observations and confirms that the inability of traditional TC to track state-dependent temperature behavior is the root cause. These findings highlight the importance of incorporating state-aware or nonlinear compensation strategies in future designs.

Based on the measurements of V_{TH} distributions and the mechanism analysis presented above, we distill three practical guidelines to improve wide temperature readability under on-chip temperature compensation in charge-trap 3D NAND:

- Programming guideline under on-chip TC: For the tested product and TC scheme, low temperature program is preferred when feasible, while programming at high temperature tends to amplify distortion in high program states with TC.
- Piecewise T_{co} behavior: We identify distinct T_{co} regimes at low and high temperatures and provide a physics-based interpretation supported by measurements and TCAD.
- High state anomaly under programming at high temperature: We observe a mean-drift inversion in P5–P7 with TC enabled that is consistently reproduced across multiple chips from the same lot, and attribute it to TC overcompensation, suggesting that high temperature program be limited to lower PV states when unavoidable.

IV. CONCLUSION

This work reveals a critical limitation in conventional on-chip TC for 3D NAND: a state-dependent overcompensation that causes abnormal mean-drift inversion in high program states (P5–P7) at elevated temperatures. We demonstrate that this anomaly arises from a mismatch between the linear TC scheme and the device's intrinsic piecewise thermal response, driven by a transition from charge stability to Poole-Frenkel emission and mobility degradation. Consequently, relying on state-independent correction is insufficient for advanced scaling. Our findings mandate a shift toward state-aware, nonlinear calibration strategies in future controllers and suggest prioritizing low-temperature programming to mitigate read-margin degradation in extreme thermal environments.

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REFERENCES

- [1] N. Righetti and G. Puzzilli, "2D vs 3D NAND technology: Reliability benchmark," 2017 IEEE International Integrated Reliability Workshop (IIRW), South Lake Tahoe, CA, USA, 2017, pp. 1-6, doi: 10.1109/IIRW.2017.8361235.
- [2] D. Resnati et al., "Characterization and Modeling of Temperature Effects in 3-D NAND Flash Arrays—Part I: Polysilicon-Induced Variability," in IEEE Transactions on Electron Devices, vol. 65, no. 8, pp. 3199-3206, Aug. 2018, doi: 10.1109/TED.2018.2838524.
- [3] G. Nicosia et al., "Characterization and Modeling of Temperature Effects in 3-D NAND Flash Arrays—Part II: Random Telegraph Noise," in IEEE Transactions on Electron Devices, vol. 65, no. 8, pp. 3207-3213, Aug. 2018, doi: 10.1109/TED.2018.2839904.
- [4] C. Zhao et al., "Investigation of Threshold Voltage Distribution Temperature Dependence in 3D NAND Flash," in IEEE Electron Device Letters, vol. 40, no. 2, pp. 204-207, Feb. 2019, doi: 10.1109/LED.2018.2886345.
- [5] C. Zambelli, L. Crippa, R. Micheloni and P. Olivo, "Cross-Temperature Effects of Program and Read Operations in 2D and 3D NAND Flash Memories," 2018 International Integrated Reliability Workshop (IIRW), South Lake Tahoe, CA, USA, 2018, pp. 1-4, doi: 10.1109/IIRW.2018.8727102.
- [6] D. Wu, H. You, X. Wang, S. Zhong and Q. Sun, "Experimental Investigation of Threshold Voltage Temperature Effect During Cross-Temperature Write-Read Operations in 3-D NAND Flash," in IEEE Journal of the Electron Devices Society, vol. 9, pp. 22-26, 2021, doi: 10.1109/JEDS.2020.3035648.
- [7] Y. Song, D. Yu, Z. Yan, Y. Wang and L. Shi, "Temperature-Aware Differential Programming for Performance and Energy Optimization on 3D NAND High-Density Flash Memory," in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 72, no. 10, pp. 5671-5682, Oct. 2025, doi: 10.1109/TCSI.2025.3551354.
- [8] M. Pesic et al., "Insights into device and material origins and physical mechanisms behind cross temperature in 3D NAND," 2023 IEEE International Reliability Physics Symposium (IRPS), Monterey, CA, USA, 2023, pp. 1-8, doi: 10.1109/IRPS48203.2023.10117898.
- [9] Y. Guo, K. Xie, X. Fang, X. Zhan, J. Wu and J. Chen, "Cross-Temperature Reliabilities in TLC 3D NAND Flash Memory: Characterization and Solution," 2023 7th IEEE Electron Devices Technology and Manufacturing Conference (EDTM), Seoul, Korea, Republic of, 2023, pp. 1-3, doi: 10.1109/EDTM55494.2023.10103107.
- [10] M. A. Kumar and B. Ray, "Cross-Temperature Reliability of 3D NAND: Cell-to-Cell Variability Analysis and Countermeasure," 2024 IEEE International Reliability Physics Symposium (IRPS), Grapevine, TX, USA, 2024, pp. P25.MR-1-P25.MR-7, doi: 10.1109/IRPS48228.2024.10529488.
- [11] Bongsik Choi et al., "Comprehensive evaluation of early retention (fast charge loss within a few seconds) characteristics in tube-type 3-D NAND flash memory," 2016 IEEE Symposium on VLSI Technology, Honolulu, HI, USA, 2016, pp. 1-2, doi: 10.1109/VLSIT.2016.7573385.
- [12] C. Woo et al., "Modeling of Charge Loss Mechanisms during the Short Term Retention Operation in 3-D NAND Flash Memories," 2019 Symposium on VLSI Technology, Kyoto, Japan, 2019, pp. T214-T215, doi: 10.23919/VLSIT.2019.8776579.
- [13] S. Xia et al., "Analysis and Optimization of Temporary Read Errors in 3D NAND Flash Memories," in IEEE Electron Device Letters, vol. 42, no. 6, pp. 820-823, June 2021, doi: 10.1109/LED.2021.3073604.
- [14] K. Lee and H. Shin, "Investigation of Retention Characteristics for Trap-Assisted Tunneling Mechanism in Sub 20-nm NAND Flash Memory," in IEEE Transactions on Device and Materials Reliability, vol. 17, no. 4, pp. 758-762, Dec. 2017, doi: 10.1109/TDMR.
- [15] H. Jo and H. Shin, "Compact Modeling of Trap-Assisted Tunneling Current in 3-D NAND Flash Memory," in IEEE Transactions on Electron Devices, vol. 72, no. 4, pp. 1745-1749, April 2025, doi: 10.1109/TED.2025.3541603.
- [16] J. Park et al., "Decomposition of Charge Loss Mechanisms in 3-D Nand Flash Memory: Impact of Cell Dimension via High-Temperature Long-Term Retention Characteristics," in IEEE Transactions on Electron Devices, vol. 71, no. 10, pp. 6040-6048, Oct. 2024, doi: 10.1109/TED.2024.3449251.
- [17] G. Malavena, M. Giulianini, L. Chiavarone, A. S. Spinelli and C. Monzio Compagnoni, "Analysis of High-Temperature Data Retention in 3D Floating-Gate nand Flash Memory Arrays," in IEEE Journal of the Electron Devices Society, vol. 11, pp. 524-530, 2023, doi: 10.1109/JEDS.2023.3320722.
- [18] K. -H. Lin et al., "A Multiscale Simulation Approach for Program Noise and Cross-temperature Effects in 3D NANDs," 2024 International Conference on Simulation of Semiconductor Processes and Devices (SISPAD), San Jose, CA, USA, 2024, pp. 1-4, doi: 10.1109/SISPAD62626.2024.10733082.